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Cyclic mechanical behavior of electro-welded meshes used as structural reinforcement

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Abstract

In this work, the monotonic and cyclic behavior of 6, 7 and 8 mm diameter electro welded meshes manufactured in Colombia and used as reinforcement in construction is studied experimentally. The cyclic mechanical behavior of the material is modeled using the Ramberg-Osgood model, and the model parameters are adjusted based on experimental data and using linear regression. The results show that the material undergoes cyclic softening, and that the Ramberg-Osgood Ramberg -Osgood model accurately predicts the monotonic and cyclic mechanical behavior.

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Keywords: Hysteresis loop; Constitutive equation; Ramberg – Osgood; Electro-Welded meshes;Cyclic behavior.

1. Introduction

Structural elements, such as reinforcing rods, beams, walls, among others, must be designed not only to support static loads but also to be able to respond adequately to cyclic loads. Although these components may have mechanical strengths that are within the ranges suggested by international standards, several studies point out that some of these components may have brittle failures associated with their low ductility when subjected to variable

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loads [1,2]. To simulate the behaviour of these materials under monotonic and cyclic loading, several analytical models available in the literature can be used, one of the most widespread being the Ramberg Osgood model [3].

The Ramberg-Osgood model has been validated in steel and aluminium metal structures, both for static and dynamic loading conditions. The analytical solutions provided by this model and its derivatives are considered valid approximations for structural component analysis [4–7]. For the case of cyclic loads, the limitations of the model in predicting hysteresis behaviour are associated with its inability to accurately predict cyclic hardening and softening of materials, as well as difficulties in characterizing buckling behaviour [8–10]

In this paper the Ramberg Osgood model will be used to characterize the behaviour of electro welded wire meshes under monotonic and cyclic loading. Electro-welded wire meshes (WWM) are widely utilized in Latin American countries as concrete reinforcement in civil infrastructure due to their cost-effectiveness and quick installation time [11–15], however, a lack of ductility has been reported in WWM used for seismic applications, which could compromise the safety of structures during shaking.

Despite extensive efforts to develop and calibrate analytical or numerical models that simulate the cyclic behaviour of ductile reinforcing steel, studies on WWM wires are limited. For this reason, this paper presents and discusses the results of an experimental study that evaluates the cyclic mechanical response of WWM manufactured in Colombia. The experimental data were adjusted using the Ramberg Osgood model. The model parameters that best describe the monotonic and cyclic elastoplastic response of the WWM are reported to help researchers and practitioners improve their inelastic models for WWM.

Nomenclature

E	Young's Modulus
K	Hardening coefficient
n	Hardening exponent
WWM	Welded wire meshes.
ε	Total strain
ε_e	Elastic component of total strain
ε_{eng}	Engineering strain
ε_p	Plastic component of total strain
σ	Stress
σ_{eng}	Engineering stress
$\Delta\varepsilon$	Change in total strain
$\Delta\sigma$	Change in total stress

2. Materials and specimens

For this investigation, specimens of different diameters manufactured in Colombia were used. These specimens were obtained from straight segments cut from WWM panels, which do not contain transverse wires to avoid welds along their length that could bias the results of the material properties. The mesh diameters used in this study are 6 mm, 7 mm, and 8 mm whose geometrical characteristics are shown in the Table. 1. All specimens comply with NTC 5806 [16] and ASTM 1064 standards [17]. It should be noted that although these standards do not specify ductility requirements, this material is commonly used as reinforcing steel in seismic regions of Colombia [2,11,17–19]

Table. 1 Geometric properties required for corrugated bars.

Bar designation	Diameter (mm)	Cross-sectional Area (mm²)
D 6.0	6	28.3
D 7.0	7	38.5
D 8.0	8	50.3

In the present work, monotonic and cyclic tests are performed using an MTS-370 servo hydraulic testing machine. Strain measurements are performed using an MTS 634.11F-24 series extensometer with a measuring gauge length of 25 mm. Monotonic tests are performed according to ASTM E8 [20] at a rate of 2 mm/min, while cyclic tests are performed according to ASTM E606 [21] at a rate of 5 mm/min using the strain history shown in figure 2.

The specimens used for the cyclic tests are deformed with positive strains, which ensures that only tensile cycles are considered to control the buckling of the bar. This decision is based on the expected asymmetry of hysteresis loops and tensile rupture commonly reported for the thin-walled structural system in laboratory and numerical studies [18,22].

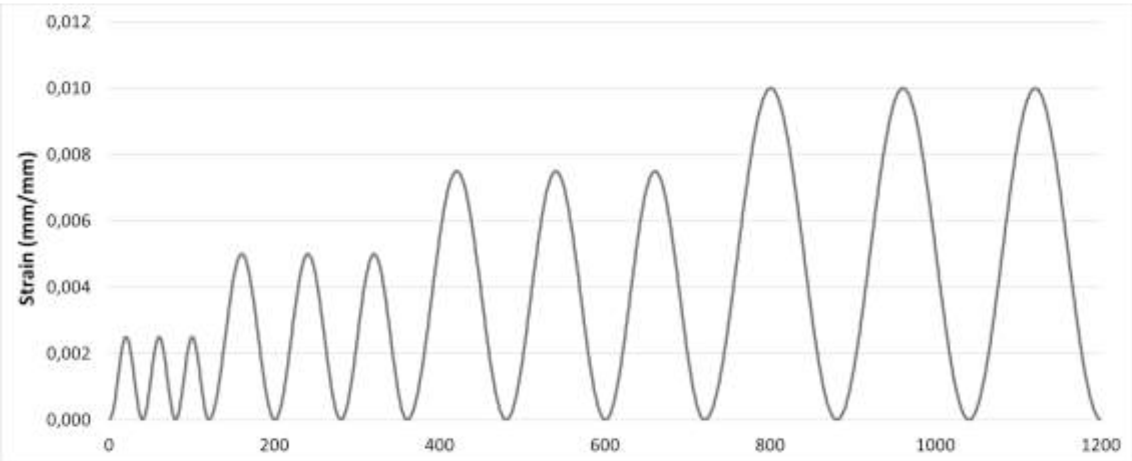

Figure 1. Strain amplitudes used for cyclic loading.

3. Ramberg Osgood model

Elastoplastic constitutive models enable simulating the mechanical response of materials, such as reinforcing steel, in both their elastic and plastic regions [23]. The Ramberg Osgood model [24] is a well-established method for modelling the stress-strain elastoplastic response of materials, as evidenced by its widespread use in the literature [25–29]. According to this model, the total strain (ε) can be divided into an elastic component and a plastic component, as follows:

$$\varepsilon = \varepsilon_\mathrm{e} + \varepsilon_\mathrm{p} \tag{1}$$

Where ε_e and ε_p represent the elastic and plastic components of the total strain, respectively. The model calculates ε by using a linear relationship for ε_e and a potential equation for ε_p, as follows:

$$\varepsilon = \left(\frac{\sigma}{E}\right) + \left(\frac{\sigma}{K}\right)^{\frac{1}{n}} \tag{2}$$

Where E is the Young's Modulus, σ is the true stress, K is the hardening coefficient, and n is the hardening exponent. This relationship is shown in Figure 2. The Ramberg Osgood model is commonly used to model the stress-strain curve under monotonic loading, but it has also been adapted to predict the cyclic response of various materials [23,27]. To simulate hysteresis loops, equation (2) is modified to an incremental form by replacing $\varepsilon = \Delta\varepsilon$, and $\sigma = \Delta\sigma$ as follows:

$$\Delta\varepsilon = \frac{\Delta\sigma}{E} + \left(\frac{\Delta\sigma}{K}\right)^{\frac{1}{n}} \tag{3}$$

Where $\Delta\sigma$ and $\Delta\varepsilon$ are the change in stress and change in strain, respectively. The parameters K and n in equation (3 are different as those used for fitting monotonic curves.

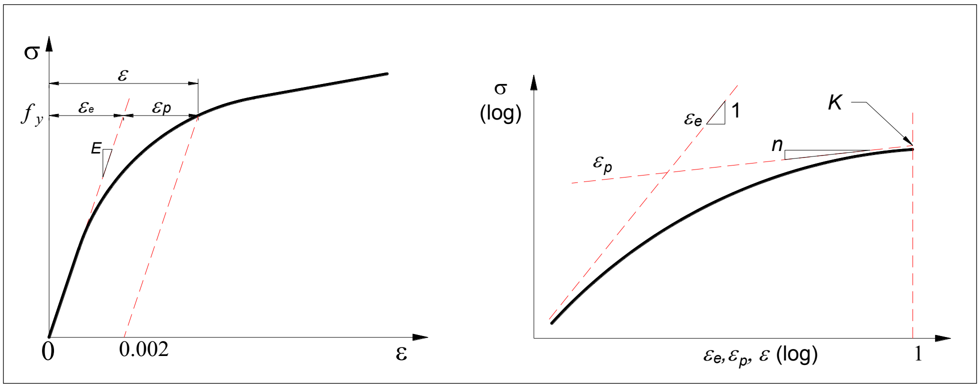

Figure 2. the Ramberg–Osgood relationship.

The Ramberg Osgood model is advantageous for its simplicity in adjusting model parameters based on experimental data. Mostaghel et al [27] and Li. J et al [28] have successfully applied the Ramberg Osgood model to characterize the hysteresis loops of different steels. However, some studies have suggested that the Ramberg Osgood model predictions are not accurate enough for certain steels [30]. Additionally, the R–O model is not suitable for predicting hysteresis loops in buckling cases during cyclic strain [31]. Likewise, it is not appropriate to simulate the cyclic hardening or softening of the materials [25].

4. Model calibration

To obtain the actual stress-strain curves from the test results the engineering stress (σ_{eng}) and strain (ε_{eng}) are computed based on the initial cross section area and gauge length Lc, respectively. Hence, σ and ε are estimated as follows [23]:

$$\sigma = \sigma_{eng}(1 + \varepsilon) \tag{4}$$
$$\varepsilon = ln(1 + \varepsilon_{eng}) \tag{5}$$

Equations (4) and (5) are applicable only until necking occurs. Hence, the various models are fitted up to f_u. To find, K and n for every test, the Ramberg-Osgood equation should be linearized.

$$\varepsilon = \frac{\sigma}{E} + \left(\frac{\sigma}{K}\right)^{\frac{1}{n}} \quad (6)$$

$$\varepsilon - \frac{\sigma}{E} = \left(\frac{\sigma}{K}\right)^{\frac{1}{n}} \quad (7)$$

$$n * ln\left(\varepsilon - \frac{\sigma}{E}\right) + ln\ (K) = ln(\sigma) \quad (8)$$

$$y = ln(\sigma) \quad (9)$$

$$x = Ln\left(\varepsilon - \frac{\sigma}{E}\right) \quad (10)$$

With these modified equations, the parameters n and K can be found by linear regression. For this model to be effective in predicting elastoplastic behaviour, the calibration must be performed in the zone where the plastic deformation is significant. One way to estimate graphically with which data it is convenient to perform the calibration of the model is to plot the stress and the plastic strain on a log-log scale, as shown in the figure 4, and observe where the curve tends to become linear. The same process is performed to calibrate the Ramberg Osgood model for hysteresis loops.

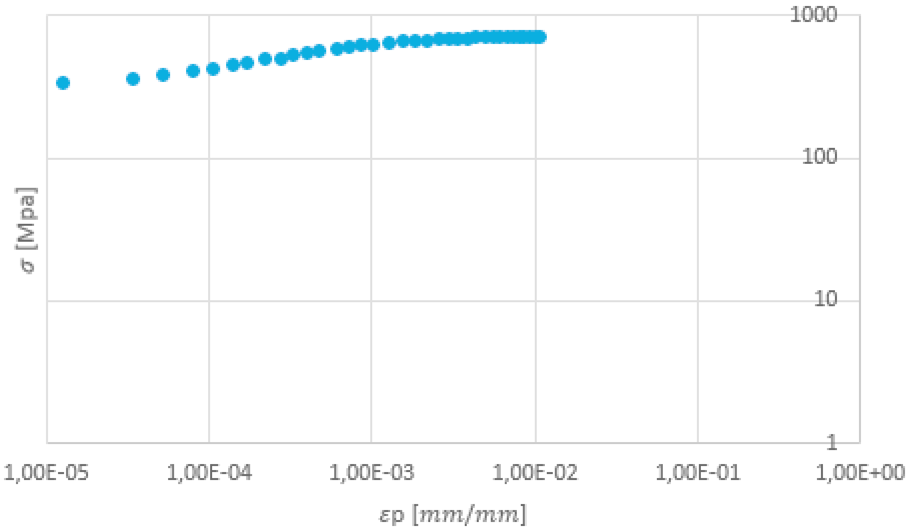

Figure 3. Stress - Plastic strain on log-log scale

5. Results and discussion

Calibration of the parameters of different elastoplastic models based on empirical data obtained from monotonic and cyclic tests can allow modelling of the elastoplastic response of individual WWM wires. This modelling capability can also be extended to reinforced concrete elements and structures reinforced with WWM, including slabs and structural walls. Figures 5-7 presents the comparison between the measured and simulated monotonic stress-strain curve using the Ramberg Osgood model for the 3 diameters tested in monotonic regime.

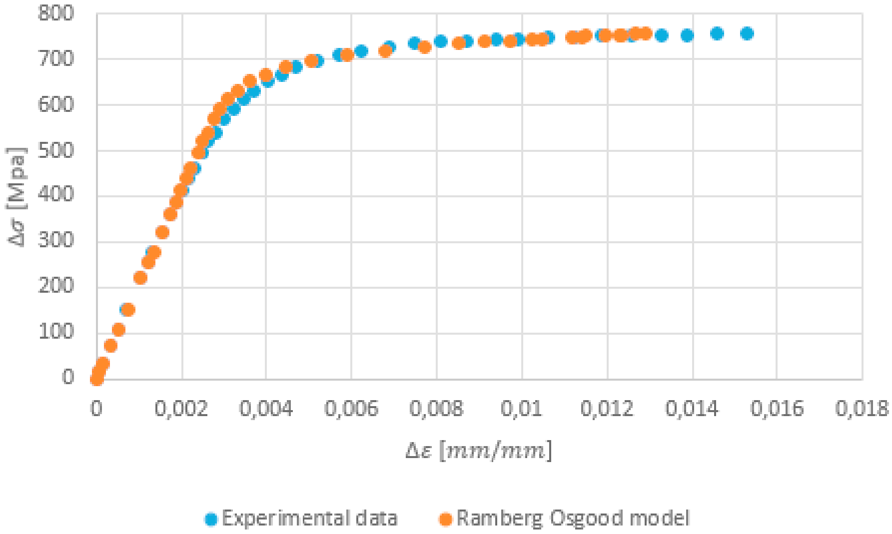

Figure 4. Monotonic curve, 6mm diameter specimen.

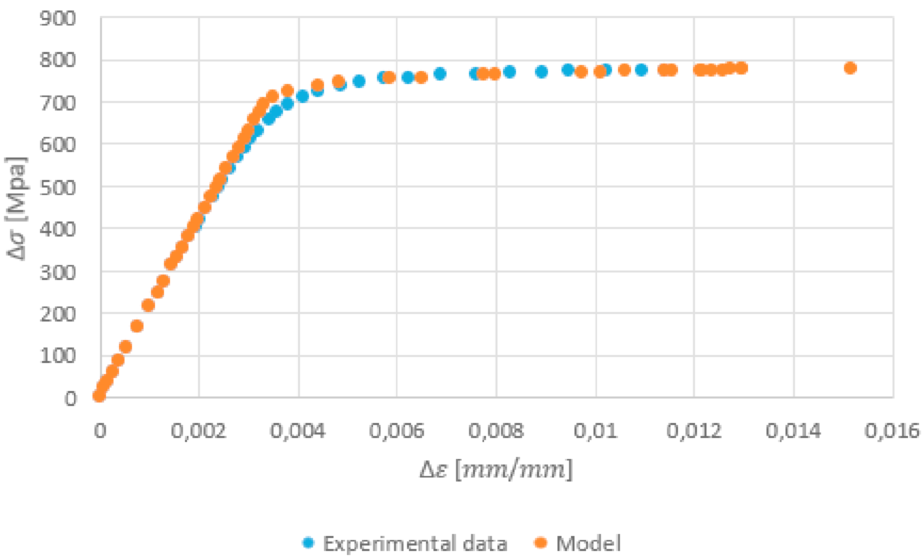

Figure 5. Monotonic curve, 7mm diameter specimen.

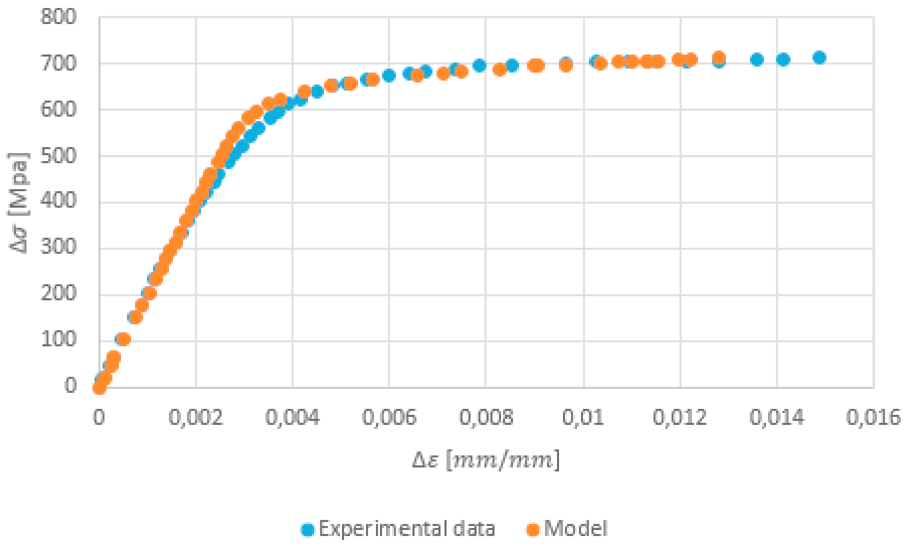

Figure 6. Monotonic curve, 8mm diameter specimen.

The parameters of the models used are shown in the Table. 2. The value of K ranges between 855Mpa and 959 Mpa, while the value of n ranges between 0.0211 -0.0506

Table. 2 Parameters of the Ramberg Osgood model for monotonic loading.

Bar designation	K [Mpa]	n
D 6.0	959	0,0506
D 7.0	855	0,0211
D 8.0	897	0,0493

The stress-strain curves obtained from the cyclic tests are shown in Figures 8 - 10. It can be observed that after each cycle, the maximum stress of the loops decreases, which is known as softening. To obtain the parameters of the Ramberg Osgood model that best fits the cyclic curve, the deltas and delta S variables must be plotted, and then follow the methodology defined in section ¡Error! No se encuentra el origen de la referencia.. T he results obtained are shown in figures 11 - 13.

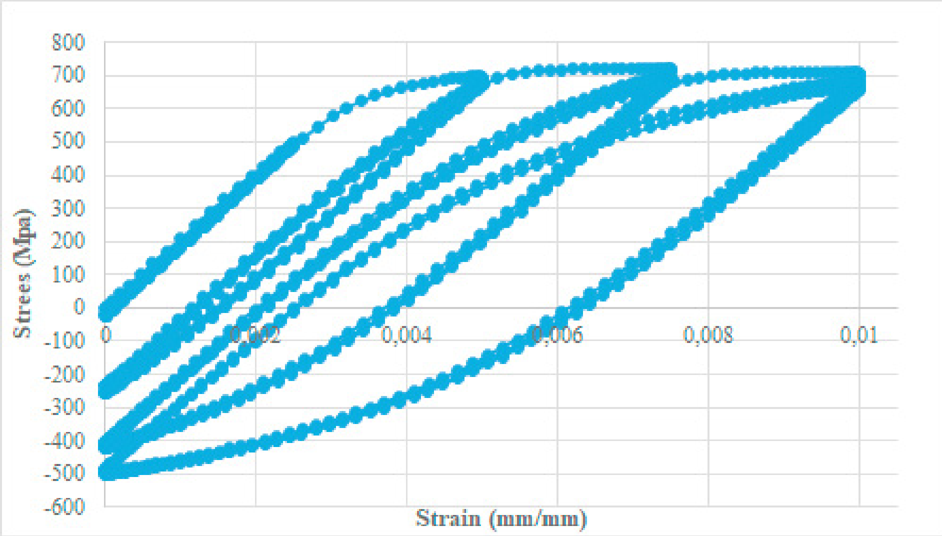

Figure 7. Cyclic curve, 6mm diameter specimen.

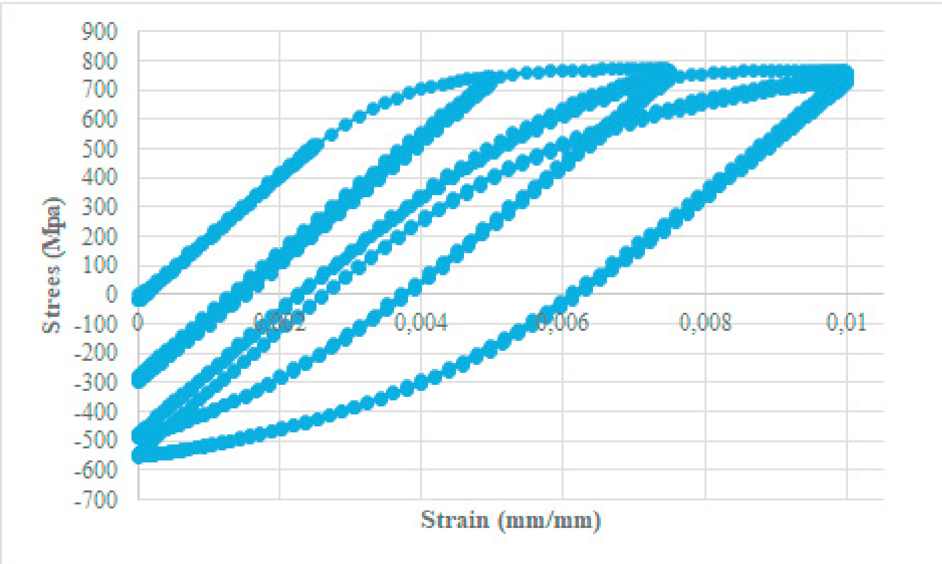

Figure 8. Cyclic curve, 7mm diameter specimen.

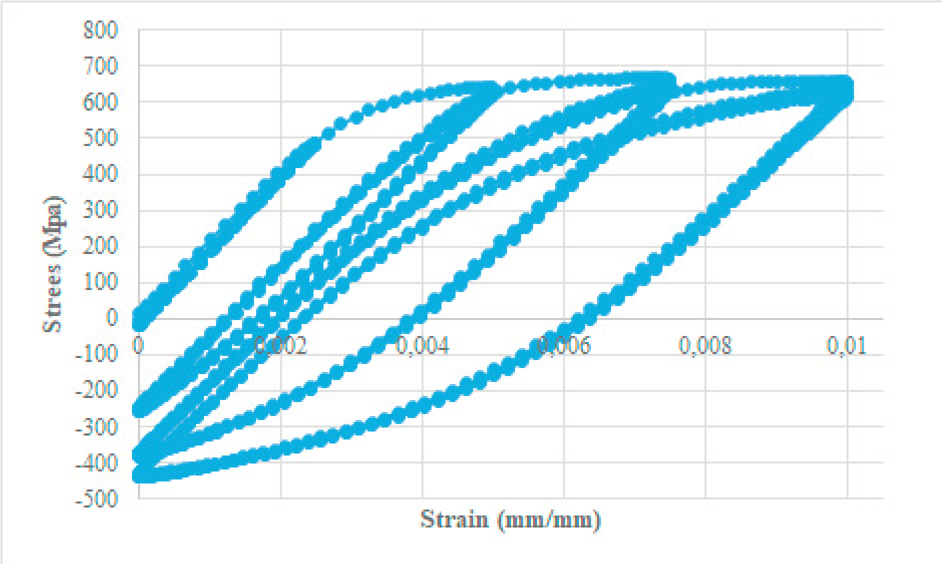

Figure 9. Cyclic curve, 8mm diameter specimen.

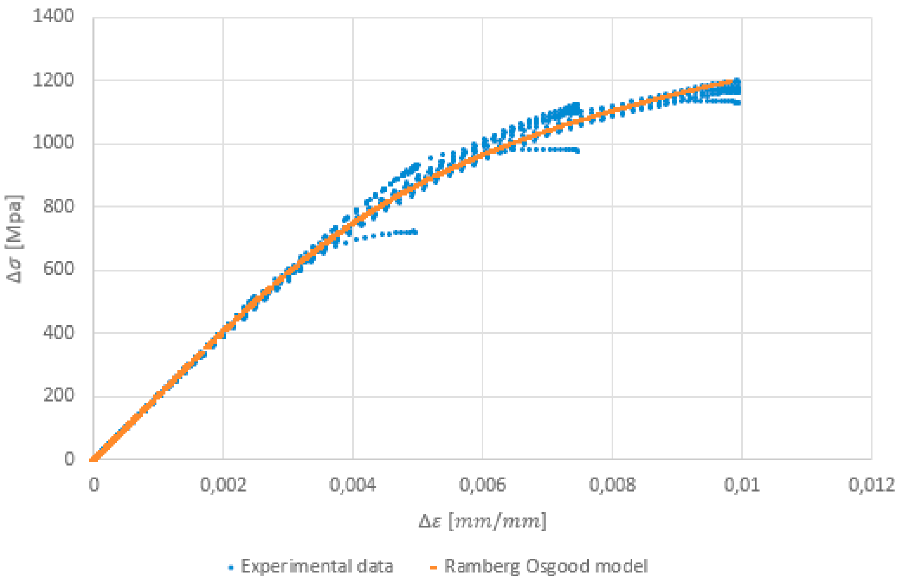

Figure 10. Cyclic curve, 8mm diameter specimen.

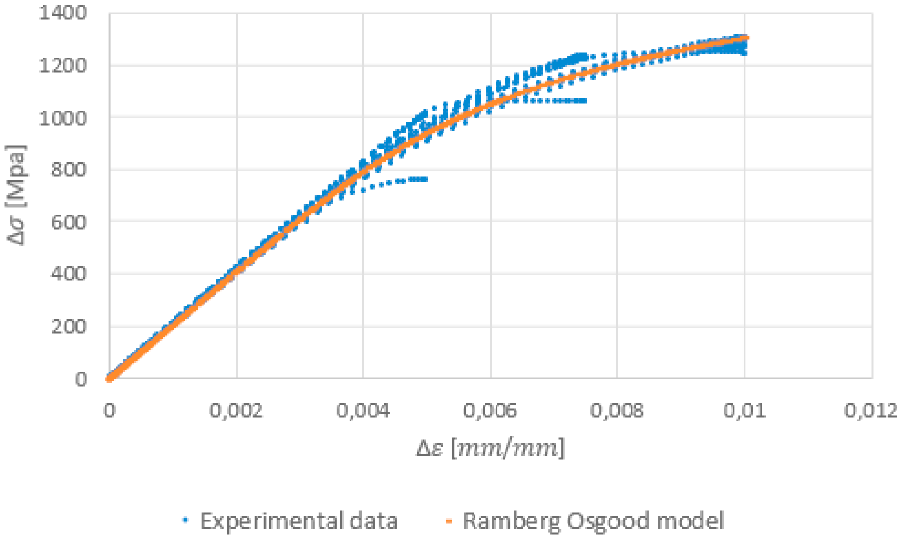

Figure 11. Cyclic curve, 8mm diameter specimen

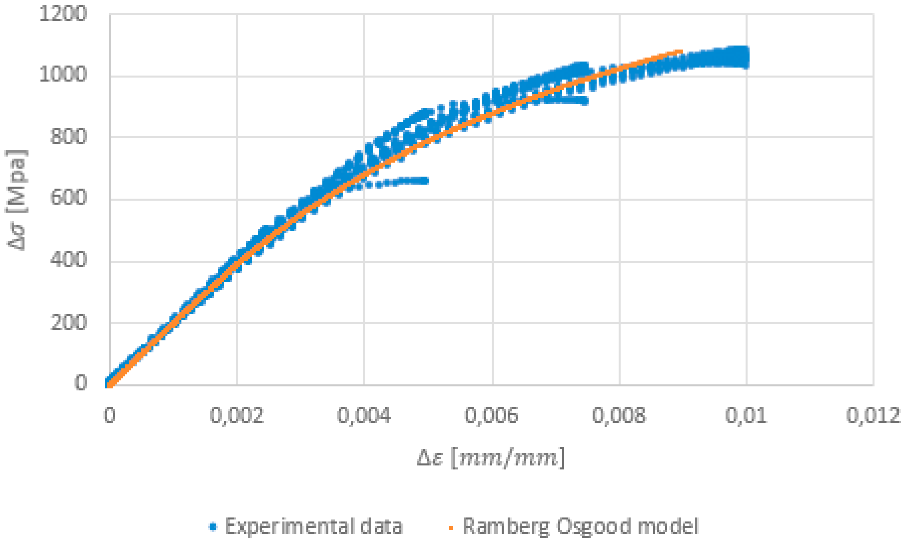

Figure 12. Cyclic curve, 8mm diameter specimen

The parameters of the models used are shown in the Table. 3.The value of K ranges between 3038 Mpa and 4851 Mpa, while the value of n ranges between 0.1504 - 0.267

Table. 3 Parameters of the Ramberg Osgood model for cyclic loading.

Bar designation	K [Mpa]	n
D 6.0	3466	0,1918
D 7.0	3038	0,1504

D 8.0	4851	0,267

6. Conclusion

In this work, the Ramberg Osgood model was used to simulate the monotonic and cyclic behaviour of welded mesh, which is an alternative reinforcement material currently used in the construction industry. Significant findings are summarized as follows:

- The mechanical cyclic behaviour of WWM indicates that this material undergoes cyclic softening, as evidenced by a drop in stress during each applied cycle. This behaviour is attributed to the hardening process that the steel undergoes during manufacturing, specifically cold plastic deformation (drawing).
- With the experimental data, Ramberg Osgood models were fitted to specimens to be tested in monotonic and cyclic tests, obtaining fits that fit the experimental data graphically. Although the models fit the data adequately, in the area near the creep limit they tend to move away from the original curve in the monotonic curve, which is a disadvantage of the model. In the case of the monotonic and cyclic curves, although the models have similar behaviours, the difference in the unit strain predicted for the same stress level can be significant as the curve approaches the ultimate stress. Therefore, it is suggested to modify this model or to lean towards alternative models to predict the behaviour of the curves more adequately.

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